FDS EXPERIMENT-10

SUBIN P – 250701750

Experiment to understand array function in Data Science.

```
[ ]  import numpy as np

[ ]  # Creating an array
     arr = np.array([10, 20, 30, 40, 50])

     print("Original Array:", arr)

     Original Array: [10 20 30 40 50]

[ ]  # 1. Sum
     print("Sum:", np.sum(arr))

     Sum: 150

[ ]  # 2. Mean
     print("Mean:", np.mean(arr))

     Mean: 30.0

[ ]  # 3. Maximum
     print("Maximum:", np.max(arr))

     Maximum: 50

[ ]  # 4. Minimum
     print("Minimum:", np.min(arr))

     Minimum: 10
```

```python
# 5. Standard Deviation
print("Standard Deviation:", np.std(arr))
```

```
Standard Deviation: 14.142135623730951
```

```python
# 6. Sorting
print("Sorted Array:", np.sort(arr))
```

```
Sorted Array: [10 20 30 40 50]
```

```python
# 7. Reshaping
arr2 = np.array([1, 2, 3, 4, 5, 6])
print("Reshaped Array:")
print(arr2.reshape(2, 3))
```

```
Reshaped Array:
[[1 2 3]
 [4 5 6]]
```

```python
# 8. Finding array size
print("Array Size:", arr.size)
```

```
Array Size: 5
```

```python
# 9. Finding dimensions
print("Number of Dimensions:", arr.ndim)
```

```
Number of Dimensions: 1
```

```python
# 10. Finding shape
print("Shape:", arr.shape)
```

```
Shape: (5,)
```

```python
# Creating a 2D array
arr2d = np.array([[1, 2, 3],
                  [4, 5, 6],
                  [7, 8, 9]])

print("\n2D Array:")
print(arr2d)
```

```
2D Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```python
# Sum of all elements
print("Sum of 2D Array:", np.sum(arr2d))
```

```
Sum of 2D Array: 45
```

```python
# Row-wise sum
print("Row-wise Sum:", np.sum(arr2d, axis=1))
```

```
Row-wise Sum: [ 6 15 24]
```

```
# Column-wise sum
print("Column-wise Sum:", np.sum(arr2d, axis=0))
```

```
Column-wise Sum: [12 15 18]
```

```
# Mean
print("Mean:", np.mean(arr2d))
```

```
Mean: 5.0
```

```
# Maximum and Minimum
print("Maximum:", np.max(arr2d))
print("Minimum:", np.min(arr2d))
```

```
Maximum: 9
Minimum: 1
```

```
# Shape
print("Shape:", arr2d.shape)
```

```
Shape: (3, 3)
```

```
# Number of dimensions
print("Dimensions:", arr2d.ndim)
```

```
Dimensions: 2
```

```
# Accessing an element
print("Element at row 2, column 3:", arr2d[1, 2])
```

```
Element at row 2, column 3: 6
```